

DR.BABASAHEBAMBEDKARTECHNOLOGICALUNIVERSITY,LONERE Winter Examination–2022 Course: B. Tech. Branch: Information Technology Semester: VII Subject Code & Name: BTITOE704A Natural Language Processing Max Marks: 60 Date:07/02/2023 Duration:3Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ()in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is speech processing in NLP? Explain use of speech processing in computing and communication.	CO5	6
B)	Explain any three applications of NLP in detail.	CO5	6
C)	Explain the phases of natural language processing application development.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Show working of NER with appropriate example.	CO2	6
B)	Write a note on morphology and its applications in NLP.	CO4	6
C)	Explain the use of regular expressions in NLP.	CO3	6
Q. 3	Solve Any Two of the following.		12
A)	What is smoothing in language modeling? Explain any two types of smoothing techniques.	CO2	6
B)	What is language modeling? Write a note on N-Gram language model.	CO1	6
C)	Explain Hidden Markov Model with suitable example.	CO1	6
Q.4	Solve Any Two of the following.		12
A)	What is POS tagging? Explain any one algorithm used for POS tagging.	CO1	6
B)	Write a note on wordnet.	CO4	6
C)	What is word sense disambiguation? Explain the use of word sense disambiguation in NLP applications.	CO2	6
Q.5	Solve Any Two of the following.		12
A)	What is pragmatics? Explain the importance of pragmatics in NLP.	CO5	6
B)	Explain the terms homonymy, polysemy, synonymy, hyponymy with examples.	CO4	6
C)	Write a note on natural language generation and its applications.	CO5	6

END